Hukou Type, Place of Work, and Experiences of Labor-Rights Problems

ONGOING LABOR RIGHTS SURVEY / RESEARCH REPORT

Comparative Analysis of Questionnaire Responses and Development of a Longitudinal Database

Survey wave	Respondents	Data coverage	Revision date
pilot-2026	59	2024-07-26 to 2026-10-02	2026-10-18

Agricultural hukou	Non-agricultural hukou	Hukou unclear
17 respondents	40 respondents	2 respondents

Key findings

At least one specific labor-rights issue was reported by 11/17 (64.7%) of respondents with agricultural hukou and 19/40 (47.5%) with non-agricultural hukou, a difference of +17.2 percentage points (95% confidence interval [CI]: [-10.6, 40.3]; two-sided P = 0.2621). The direction of the difference is consistent with disadvantage for the agricultural group, but the estimate lacks precision.

Differences were +33.7 percentage points for wage arrears and +23.5 points for work injury or occupational-safety problems. Their P values after Holm correction across eight items were 0.0393 and 0.0422. Item-specific outcomes reveal more than the combined outcome, but are also more sensitive to how questionnaire responses are processed.

Conclusions are grounded in observed sample data and statistical uncertainty. A failure to detect differences within the non-agricultural group does not establish that those factors have no effects. Observed associations are not presented as independent causal effects of the hukou system.

“Agricultural hukou” and “non-agricultural hukou” follow the questionnaire categories. Hukou type is analyzed separately from current residence and place of work.

01 / Study Objectives and Analytical Methods

This study examines past reports of labor-rights problems among people with different hukou classifications. It focuses on specific experiences showing relatively large differences between agricultural and non-agricultural hukou, and examines how those differences appear when stratified by place of work, age, industry, income and employment arrangement. The data come from a cross-sectional convenience-sample questionnaire, not a random sample of the overall workforce.

Analytical level	Outcomes and comparisons	Scope of interpretation
Primary comparison	Two hukou groups; at least one specific Q21 issue	Estimate within-sample difference in experience prevalence
Specific rights issues	Q21 items 1-8; report all eight	Holm correction within each comparison family
Place of work	Prespecified main rule and strict explicit-response rule	Distinguish hukou status from the proxy for living/working outside the registered locality
Within non-agricultural hukou	Age, industry, income, employment and location	Examine within-group differences and uncertainty
Robustness checks	Exclude contradictory, explicitly non-working and underage responses	Assess whether a small number of responses materially alter results

How to read the statistics

First, count the respondents in each group and how many reported an issue; then calculate proportions and the between-group difference. In a table, “7/17” means that 7 of 17 respondents selected an item. Differences are agricultural minus non-agricultural, or non-local minus local, in percentage points. Fisher’s exact test is used for two-group comparisons because sample sizes are small [4].

Confidence intervals show uncertainty in estimates from small samples. Intervals for a single proportion use the Wilson method, while intervals for the difference between two proportions use the Newcombe method [2]. When eight rights issues are compared simultaneously, Holm adjustment [3] reduces false positives arising from multiple testing. Each interval is computed separately; the whole table does not have simultaneous 95% coverage.

Calculations were performed in Python; the code and algorithms are retained in the companion project. Comparisons involving several groups, such as industries, additionally use a permutation test (19,999 permutations; seed 20261013). This is an exploratory analysis of existing questionnaire data; not all supplementary comparisons received a common multiplicity adjustment. A small P value does not resolve sampling bias.

The questionnaire and supplementary coding are the data sources [1]. The final page cites the statistical methods actually used, so that calculations can be checked.

02 / Data Scope and Coding Quality

There are 59 responses. Hukou classification permits an explicit two-group comparison for 57 responses; the remaining responses are retained in full-sample and place-of-work analyses, but treated separately in hukou comparisons. Original response options, open-ended text and derived variables are stored separately; derived variables do not overwrite original answers.

Data-quality flag	Count	Main handling
Location unclear; main rule assigns non-local, strict rule excludes	33	Assign non-local under the prespecified rule; retain “unknown” flag
Industry ambiguous; not forcibly inferred	6	Retain original value; mark missing or exclude in sensitivity analysis if necessary
Daily hours cannot be parsed or exceed 16	12	Retain original value; use missingness/exclusion analysis as needed
Weekly days cannot be parsed as explicit numbers/ranges	5	Retain original value; use missingness/exclusion analysis as needed
Specific issue and “never experienced” both selected; main rule prioritizes specific issue	2	Retain original value; use missingness/exclusion analysis as needed
“Took no action” and an action option both selected	2	Retain original value; use missingness/exclusion analysis as needed

A response may carry multiple quality flags, so the counts cannot be added to obtain the number of problematic responses. When a specific issue conflicts with “never experienced,” the main analysis retains the specific experience; results excluding contradictory responses are shown separately. Industries that cannot be confirmed are not assigned to a category.

Questionnaire limitations that must be retained

Q21 asks whether an issue was experienced “in the past,” without a common recall period. Salary, industry and hours questions mostly concern current circumstances, which may refer to a different job. Explicit student, unemployed and retired responses remain in the main table of historical experiences but are omitted from relevant sensitivity samples.

Q24 has 12 options in the original questionnaire; option 12 is “I did not know my rights had been violated” and is retained. Although Q13 and Q28 contain values from 0 to 5, endpoints were not provided; they cannot support a standardized rights-awareness score or intensity of hukou influence. The spreadsheet’s “total score” is undefined and excluded from analysis.

In Q6, outsourcing, temporary, uncertain and other arrangements are combined with dispatch work in a second group under the prespecified coding rule, labeled only the “broad informal group.” It is not equivalent to labor dispatch as legally defined. “Prefer not to answer” in Q12 is a missing-income category, not zero income.

03 / Composition of the Two Hukou Groups

Non-agricultural hukou accounts for 67.8% of all respondents and agricultural hukou for 28.8%. Unequal group sizes produce wider intervals for the agricultural group, but discarding non-agricultural observations would neither add information about the agricultural group nor eliminate selection bias.

Industry grouping	All	Agricultural	Non-agricultural	Unclear
Other services and commerce	24	8	16	0
Healthcare, education and public sector	4	1	3	0
Not currently working	2	0	2	0
Production, construction and transport	9	2	6	1
Industry unclear	6	3	2	1
Design and technical professions	14	3	11	0

Employment category	All	Agricultural	Non-agricultural	Unclear
Broad informal group	22	8	13	1
Formal employees	37	9	27	1

Industry, age distributions and income nonresponse rates may differ. If those variables are associated with both hukou and experiences of rights problems, they may confound the observed relationship; a non-significant sample test does not rule this out. Current industry and income may also lie on a pathway through which hukou affects job opportunities; mechanically controlling for them changes the research question.

This wave provides no sampling frame, individual inclusion probabilities, regional population benchmarks or recruitment-network records. The analysis therefore does not use population weighting, inverse-probability weighting or forced balancing to manufacture representativeness. Stratified comparisons are shown only where both hukou types occur within a stratum.

Respondents from the same workplace or social network may be correlated, so the assumption that questionnaire responses are independent remains unverified. Future waves should record anonymized workplace/recruitment-group identifiers and survey sites, distinguishing individual associations from workplace conditions and geographic differences.

04 / Experiences of Rights Problems by Hukou Type

Reported experience	Agricultural	Non-agric.	Difference (pp) 95% CI	P	Holm P
At least one specific rights issue	11/17 (64.7%)	19/40 (47.5%)	+17.2 [-10.6, 40.3]	0.2621	-
Unpaid wages / wage arrears	7/17 (41.2%)	3/40 (7.5%)	+33.7 [10.5, 57.0]	0.0049	0.0393
Overtime without corresponding pay	8/17 (47.1%)	11/40 (27.5%)	+19.6 [-6.4, 44.3]	0.2199	0.7186
No employment contract signed	4/17 (23.5%)	3/40 (7.5%)	+16.0 [-2.6, 40.3]	0.1797	0.7186
Social-insurance problems	5/17 (29.4%)	4/40 (10.0%)	+19.4 [-1.3, 43.9]	0.1086	0.5432
Charges the worker should not bear	3/17 (17.6%)	1/40 (2.5%)	+15.1 [-0.3, 38.6]	0.0749	0.4493
Work injury or occupational safety	4/17 (23.5%)	0/40 (0.0%)	+23.5 [7.0, 47.3]	0.0060	0.0422
Unlawful termination of employment	1/17 (5.9%)	1/40 (2.5%)	+3.4 [-8.1, 24.6]	0.5113	0.9284
Labor-dispatch problems	4/17 (23.5%)	6/40 (15.0%)	+8.5 [-11.3, 33.6]	0.4642	0.9284

The primary outcome is at least one specific rights issue. The eight issue types are not mutually exclusive; a respondent could select several, so the percentages cannot be added to obtain an overall prevalence. The P value for the primary outcome is not included in the eight-item multiplicity adjustment.

The overall difference is +17.2 percentage points and its 95% interval crosses zero. The interval permits both a slightly lower rate in the agricultural group and a substantial disadvantage for that group. A P value above 0.05 cannot by itself establish that no difference exists.

The agricultural group reported a mean of 2.12 categories of issues and the non-agricultural group 0.72. The number of categories indicates co-occurrence, not the number of incidents, amount of loss or severity; it is not a validated rights index.

Items such as “no employment contract” and “work injury/occupational safety” are self-reports, not independently verified legal determinations. Accordingly, the report uses the statistical description “experiences of rights problems.”

05 / Specific Issues with Larger Differences

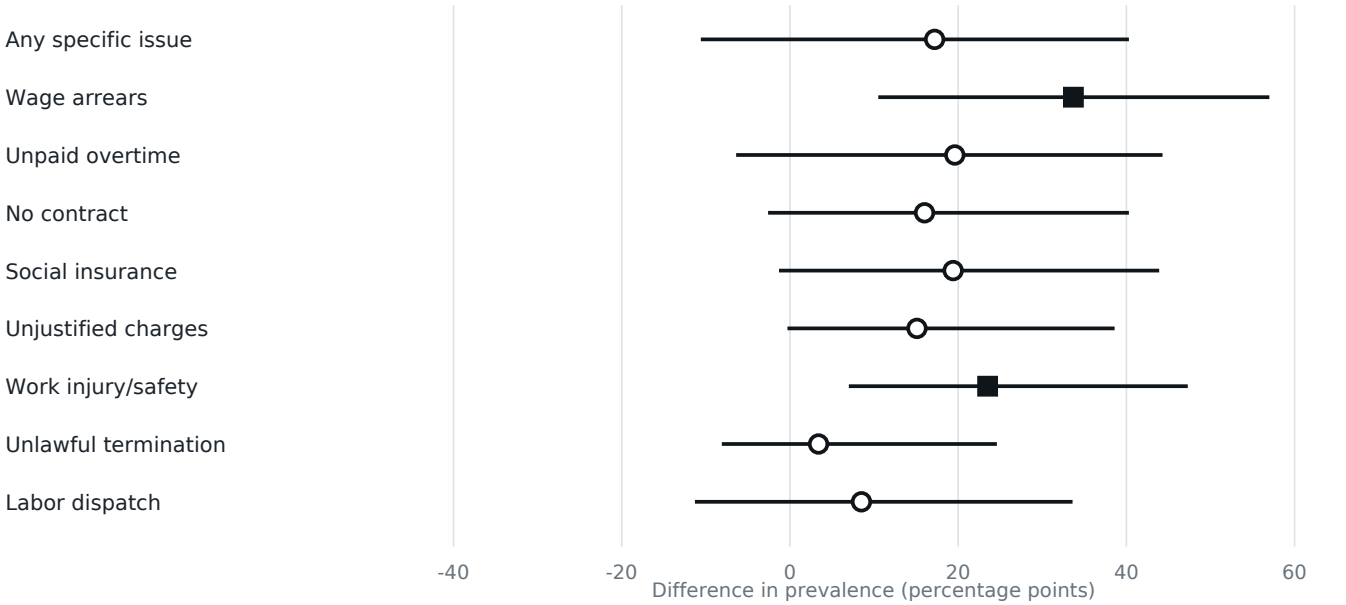


Chart: agricultural minus non-agricultural prevalence differences and 95% confidence intervals. Markers represent estimates and lines the intervals. Solid squares identify items with Holm-adjusted P < 0.05 across the eight specific outcomes; others are open circles.

Wage arrears

Agricultural: 7/17 (41.2%); non-agricultural: 3/40 (7.5%). Difference: +33.7 percentage points; 95% CI [10.5, 57.0]; sample risk ratio 5.49. This concerns whether earned income is actually paid. Future surveys should record the duration and amount of arrears, whether payment was recovered, and the type of employer when the event occurred.

Work injury and occupational safety

Agricultural: 4/17 (23.5%); non-agricultural: 0/40 (0.0%). Difference: +23.5 percentage points; 95% CI [7.0, 47.3]. Zero reports in the non-agricultural group do not mean that underlying risk is zero. The zero cell prevents a finite conventional risk-ratio estimate. The discussion therefore prioritizes the proportion difference, rather than adding an arbitrary 0.5 to manufacture a finite ratio.

Cost-shifting, missing contracts, social insurance and overtime pay also merit follow-up. However, adjusted results in the main table do not support calling each a robust difference. Reporting only the smallest unadjusted P values is inappropriate, and a large point difference is not necessarily precisely estimated.

06 / Non-local vs Local: Prespecified Main Rule

Seventeen respondents explicitly indicated “local,” nine explicitly indicated “non-local,” and 33 were unclear. Under the prespecified rule, the last two categories are combined into the non-local group; only responses explicitly denying non-local residence or explicitly identifying the hukou locality are classified as local. Years, place names, or “born in a city” alone do not establish local/non-local status.

Reported experience	Non-local (main)	Local	Difference (pp) 95% CI	P	Holm P
At least one specific rights issue	23/42 (54.8%)	9/17 (52.9%)	+1.8 [-23.8, 27.9]	1.0000	-
Unpaid wages / wage arrears	9/42 (21.4%)	3/17 (17.6%)	+3.8 [-21.5, 22.3]	1.0000	1.0000
Overtime without corresponding pay	15/42 (35.7%)	5/17 (29.4%)	+6.3 [-20.6, 28.4]	0.7659	1.0000
No employment contract signed	6/42 (14.3%)	2/17 (11.8%)	+2.5 [-21.3, 18.5]	1.0000	1.0000
Social-insurance problems	6/42 (14.3%)	4/17 (23.5%)	-9.2 [-34.2, 10.2]	0.4525	1.0000
Charges the worker should not bear	4/42 (9.5%)	1/17 (5.9%)	+3.6 [-18.2, 17.1]	1.0000	1.0000
Work injury or occupational safety	2/42 (4.8%)	2/17 (11.8%)	-7.0 [-29.8, 6.9]	0.5712	1.0000
Unlawful termination of employment	2/42 (4.8%)	0/17 (0.0%)	+4.8 [-14.0, 15.8]	1.0000	1.0000
Labor-dispatch problems	9/42 (21.4%)	2/17 (11.8%)	+9.7 [-14.9, 26.5]	0.4838	1.0000

Under the main rule, at least one issue was reported by 23/42 (54.8%) non-local and 9/17 (52.9%) local respondents; difference +1.8 percentage points, 95% CI [-23.8, 27.9]. Although the point estimates are close, the interval still permits meaningful differences, so the effect of work location cannot be dismissed.

Q3 jointly asks about long-term residence outside the hukou locality and how long the respondent has worked in the current city. It does not directly verify the workplace or administrative boundary. This section therefore compares a questionnaire-derived residence/work-location proxy, not person-by-person matches of official hukou locality with actual workplace.

07 / Location-Coding Sensitivity and Hukou Cross-Tabulation

Analysis rule	Non-local	Local	Difference (pp) / 95% CI	P
Main: unclear assigned non-local	23/42 (54.8%)	9/17 (52.9%)	+1.8 [-23.8, 27.9]	1.0000
Strict: explicit answers only	7/9 (77.8%)	9/17 (52.9%)	+24.8 [-13.8, 52.0]	0.3989

When only explicit answers are compared, the difference in overall experience prevalence rises to +24.8 percentage points. The strict rule reduces inferred classifications but further shrinks the sample; explicit respondents may not represent all non-local workers. Together, the two results show that classification uncertainty is a major limitation of the location analysis.

Hukou and work location are not the same variable

Hukou type	Explicitly local	Explicitly non-local	Unclear
Agricultural	5	4	8
Non-agricultural	12	5	23
Unclear	0	0	2

Within same location stratum	Agricultural	Non-agricultural	Difference (pp) / 95% CI
Local	4/5 (80.0%)	5/12 (41.7%)	+38.3 [-11.6, 66.0]
Non-local (including unclear)	7/12 (58.3%)	14/28 (50.0%)	+8.3 [-23.3, 36.6]

Stratification by location under the main rule avoids equating every person with agricultural hukou with migrant employment. However, the within-stratum samples are too small to distinguish the roles of hukou type, migration experience, employer, industry or position. Future questionnaires should ask separately about hukou type, registered locality, usual residence and actual workplace.

08 / Age Structure and Within-Age Comparisons

Age group	All	Agricultural	Non-agricultural	Unclear
18-25	21	9	11	1
Under 18	2	1	1	0
26-30	6	1	5	0
31-40	20	5	14	1
41-50	3	1	2	0
51-60	4	0	4	0
Over 60	3	0	3	0

All seven original age bands are retained. Because older age bands are small, display groups are also combined as 25 or younger, 26-40, and 41 or older; the binary contrast uses 30 or younger versus 31 or older. The same aggregation rules apply to both hukou types and were not chosen post hoc based on outcomes.

Within non-agricultural hukou: three age groups

Age group	N	Reported issue	Prevalence	95% CI
25 or younger	12	4/12	33.3%	[13.8, 60.9]
26-40	19	11/19	57.9%	[36.3, 76.9]
41 or older	9	4/9	44.4%	[18.9, 73.3]

The binary comparison is 8/17 (47.1%) versus 11/23 (47.8%), a difference of -0.8 percentage points. Although the point estimates are close, the 90% interval [-25.2, 24.0] is still too wide to fit within the illustrative equivalence margin of +/-10 percentage points.

Q21 has no fixed recall period, so greater age may simply mean longer exposure to opportunities for a past event. These data cannot estimate incidents per year worked. Future surveys should standardize a past-12-month period and separately record lifetime experiences and recent events. Sensitivity results excluding respondents under 18 are presented later.

09 / Industry Composition and Within-Industry Comparisons

Industry group	N	Reported issue	Prevalence	95% CI
Other services and commerce	24	13/24	54.2%	[35.1, 72.1]
Healthcare, education and public sector	4	2/4	50.0%	[15.0, 85.0]
Production, construction and transport	9	5/9	55.6%	[26.7, 81.1]
Design and technical professions	14	7/14	50.0%	[26.8, 73.2]

Industries were grouped from open-ended responses using a prespecified dictionary. “Architectural design” belongs in design/professional services, while generic “construction” is grouped with production, construction and transport; healthcare/education/public-sector and other services/commerce are separate. Insufficiently specified “technology,” “engineering,” “ordinary worker,” and unclear titles are not forcibly classified. These coarse groupings serve this exploratory comparison and are not official industry classifications.

Within non-agricultural hukou: known industries

Industry group	N	Reported issue	Prevalence	95% CI
Other services and commerce	16	8/16	50.0%	[28.0, 72.0]
Healthcare, education and public sector	3	2/3	66.7%	[20.8, 93.9]
Production, construction and transport	6	2/6	33.3%	[9.7, 70.0]
Design and technical professions	11	5/11	45.5%	[21.3, 72.0]

Within this group, prevalence was 5/11 (45.5%) in design/professional services and 8/16 (50.0%) in other services/commerce: difference -4.5 percentage points, 95% CI [-37.2, 29.9]. The overall four-industry permutation test gave P = 0.8828 (adjusted P = 1.0000).

A failure to detect overall differences means only that this sample provides insufficient discrimination. Categories are broad, some cells very small, and occupational hazards and employer conditions can vary greatly within an industry. It does not establish that industry causes no differences in employment conditions.

10 / Income Distribution and Interpreting Wage Differences

Income band (CNY)	All	Agricultural	Non-agricultural	Unclear
1,000-1,999	3	1	2	0
10,000 or more	2	0	2	0
2,000-2,999	8	3	5	0
3,000-4,999	27	7	19	1
5,000-6,999	5	2	3	0
7,000-9,999	4	1	3	0
Prefer not to answer	9	3	5	1
Below 1,000	1	0	1	0

Income is reported in ranges, not as a continuous monetary amount. The highest bracket is open-ended; “prefer not to answer” is excluded from valid income categories. To avoid creating a spurious precise mean through assigned midpoints, this report does not compute nominal average earnings or treat category codes 1-7 as actual renminbi. The primary exploratory threshold is CNY 5,000.

Within non-agricultural hukou: two valid income groups

Income group	N	Reported issue	Prevalence	95% CI
CNY 5,000 or above	8	2/8	25.0%	[7.1, 59.1]
Below CNY 5,000	27	14/27	51.9%	[34.0, 69.3]

In the below-CNY 5,000 group, 14/27 (51.9%) reported an issue, compared with 2/8 (25.0%) at CNY 5,000 or above: a difference of +26.9 percentage points. Two-sided P = 0.2438 is above the conventional threshold, but the point difference is not small. The 95% interval [-11.6, 51.8] cannot rule out a substantial income-related difference.

Current income may reflect job differences, but may also be affected by wage arrears, overtime hours or rights problems. The timing of income relative to past events is unknown, so a residual hukou association after adjustment for income cannot automatically be called the “net effect” of hukou. Future surveys should measure contracted and actual wages before the event, the arrears amount, and hours at that time.

11 / Employment Arrangements, Hours and Co-occurring Issues

Within non-agricultural hukou: employment arrangement

Group	N	Reported issue	Prevalence	95% CI
Broad informal group	13	6/13	46.2%	[23.2, 70.9]
Formal employees	27	13/27	48.1%	[30.7, 66.0]

The primary-outcome difference between the broad informal group and formal employees is -2.0 percentage points (95% CI [-31.1, 28.2]). The combined category includes labor dispatch, outsourcing, temporary, uncertain and other arrangements, with potentially substantial internal heterogeneity. These data do not show that legally defined dispatched workers receive the same treatment as formal employees.

Parseable working hours and number of specific issue categories

Group	N with valid hours	Median weekly hours	Interquartile range	Mean issue categories
All	44	44.5	40.0-55.5	1.22
Agricultural	13	48.0	45.0-70.0	2.12
Non-agricultural	30	40.0	40.0-50.8	0.72

Weekly hours were estimated as explicitly stated days per week multiplied by hours per day; interval answers use the midpoint. Ambiguous day/week units, vague time descriptions and days exceeding 16 hours are not given quantitative values, although original answers are retained. Explicit non-workers are excluded from current hours. These descriptive figures do not subtract breaks or verify special working-hour systems.

Fewer valid hours are available for the agricultural group, and missingness may be associated with work arrangements. The differences in medians and average issue-category counts identify topics worth measuring, but they are not combined into an unvalidated “overall employment-treatment score,” and daily hours are not used alone to make legal determinations.

A comprehensive description of employment conditions also requires actual pay, employer contributions, written contracts, rest days, occupational protection and the outcomes of dispute resolution. This questionnaire focuses more narrowly on rights-related experiences.

12 / Are Differences Small Within Non-agricultural Hukou?

Comparison (first minus second)	First / second prevalence	Difference (pp)	90% CI	Within +/-10 pp?
Age: 30 or younger / 31 or older	8/17 (47.1%) 11/23 (47.8%)	-0.8	[-25.2, 24.0]	Insufficient
Income: below 5,000 / 5,000 or above	14/27 (51.9%) 2/8 (25.0%)	+26.9	[-5.9, 49.0]	Insufficient
Industry: design/technical / other services/commerce	5/11 (45.5%) 8/16 (50.0%)	-4.5	[-33.1, 25.3]	Insufficient
Employment: broad informal / formal	6/13 (46.2%) 13/27 (48.1%)	-2.0	[-27.2, 24.1]	Insufficient
Location: non-local (incl. unclear) / local	14/28 (50.0%) 5/12 (41.7%)	+8.3	[-18.8, 33.0]	Insufficient

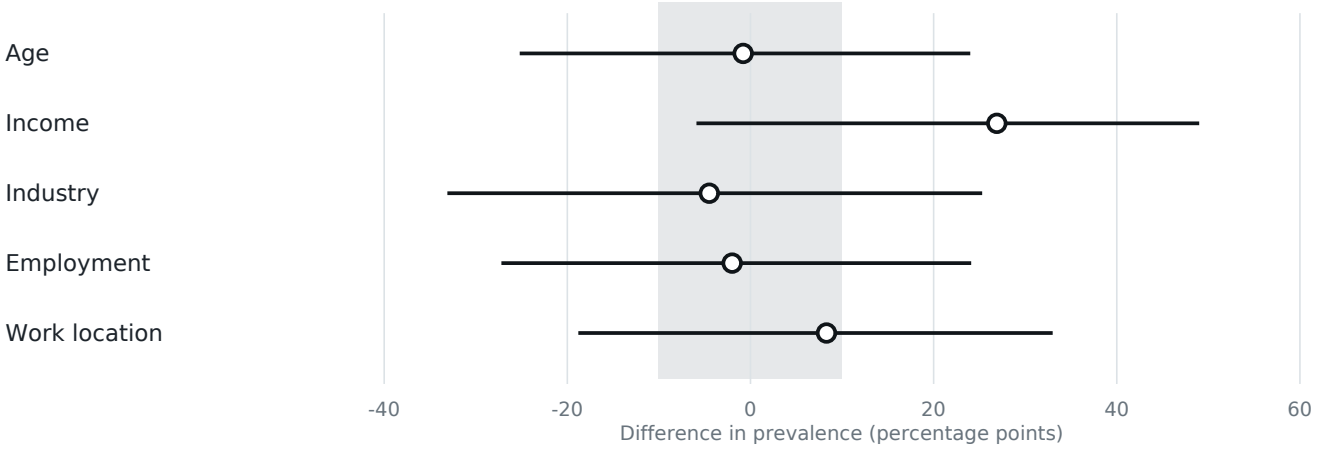


Chart: the shaded region is the illustrative equivalence margin of +/-10 percentage points; lines show 90% intervals. The margin illustrates the distinction between “no difference detected” and “sufficiently close,” not an externally validated policy threshold.

None of the five binary comparisons has an interval entirely inside the illustrative equivalence range. An interval extending outside the limits does not provide sufficient evidence that a difference is “very small.” Point estimates near zero or large P values do not justify saying these factors cannot lead to differences in rights experiences.

All comparisons involve observational subgroups within the same convenience sample, not randomly assigned control groups. Even if smaller within-non-agricultural differences are precisely supported in future, that alone would not prove that differences for agricultural hukou are caused by the hukou system; unmeasured differences in occupations and regions may remain.

13 / Do Results Change Under Alternative Processing Rules?

Sample rule	Agricultural	Non-agricultural	Primary difference (pp) / 95% CI	P
Main analysis	11/17 (64.7%)	19/40 (47.5%)	+17.2 [-10.6, 40.3]	0.2621
Exclude contradictory Q21 responses	10/16 (62.5%)	18/39 (46.2%)	+16.3 [-12.0, 40.3]	0.3753
Exclude explicitly not working	11/17 (64.7%)	19/37 (51.4%)	+13.4 [-14.6, 37.1]	0.3940
Exclude respondents under 18	11/16 (68.8%)	18/39 (46.2%)	+22.6 [-6.1, 45.1]	0.1494
Exclude contradictions, non-workers and minors	10/15 (66.7%)	17/35 (48.6%)	+18.1 [-11.5, 42.0]	0.3545

Adjusted multiple-comparison results for key items

Sample rule	Wage arrears: difference / Holm P	Work injury/safety: difference / Holm P
Main analysis	+33.7 / 0.0393	+23.5 / 0.0422
Exclude contradictory Q21 responses	+29.8 / 0.1034	+18.8 / 0.1494
Exclude explicitly not working	+33.1 / 0.0562	+23.5 / 0.0562
Exclude respondents under 18	+38.6 / 0.0113	+25.0 / 0.0374
Exclude contradictions, non-workers and minors	+34.3 / 0.0478	+20.0 / 0.1625

Excluding conflicting responses changes the small number of rare events; excluding non-workers also changes the denominator in the non-agricultural group. Statistical significance changes across rules, indicating limited robustness; only the most significant specification must not be selected. Similar directions and statistical significance are different concepts, and agreement in direction cannot replace reporting uncertainty.

Extreme assignment of the two unclear-hukou responses is considered only as a sensitivity scenario: assigning both to agricultural hukou yields a primary-outcome difference of +20.9 percentage points; assigning both to non-agricultural hukou yields +14.7 points. Neither scenario changes the hukou coding in the source database or asserts the respondents' true status.

14 / Rights Awareness and Willingness to Follow Regulations

Rights respondent says they know	Agricultural	Non-agricultural	Difference (pp) / 95% CI	Holm P
Minimum wage	14/17 (82.4%)	31/40 (77.5%)	+4.9 [-20.7, 23.7]	1.0000
Overtime pay	15/17 (88.2%)	35/40 (87.5%)	+0.7 [-22.9, 16.8]	1.0000
Social insurance	14/17 (82.4%)	36/40 (90.0%)	-7.6 [-31.8, 9.7]	1.0000
Employment contract	17/17 (100.0%)	39/40 (97.5%)	+2.5 [-16.0, 12.9]	1.0000
Labor-dispute arbitration	15/17 (88.2%)	35/40 (87.5%)	+0.7 [-22.9, 16.8]	1.0000
Work-injury-related rights	15/17 (88.2%)	33/40 (82.5%)	+5.7 [-18.5, 22.5]	1.0000

These responses measure self-reported awareness, not tested legal knowledge, and do not directly measure how fully rights are realized. A descriptive count across six items gives mean numbers of known rights of 5.29 and 5.22 in agricultural and non-agricultural groups, respectively. Knowing one’s rights and having them protected may differ considerably.

Expected income and choices if regulations are strictly followed

Response option	Agricultural N	Non-agricultural N
Expected income: essentially unchanged	5	13
Expected income: increases	6	11
Expected income: decreases	1	4
Expected income: uncertain	5	12
Choice: insist on full compliance	11	20
Choice: accept some of the status quo	4	19
Choice: depends on circumstances	2	1

These responses express attitudes to a hypothetical scenario, not observed income changes or dispute outcomes. Attitude differences do not imply what employment conditions a worker should accept, and similar rights awareness does not imply the absence of other resource constraints.

15 / Responses After Experiencing Rights Problems

Response option	Agricultural	Non-agricultural	Difference (pp) / 95% CI
Negotiated with boss/supervisor/team leader	5/11 (45.5%)	9/19 (47.4%)	-1.9 [-33.9, 31.3]
Contacted HR	3/11 (27.3%)	5/19 (26.3%)	+1.0 [-27.5, 33.6]
Contacted a trade union	1/11 (9.1%)	2/19 (10.5%)	-1.4 [-23.6, 28.2]
Labor inspection	2/11 (18.2%)	2/19 (10.5%)	+7.7 [-17.0, 38.1]
Labor arbitration	3/11 (27.3%)	1/19 (5.3%)	+22.0 [-4.1, 51.6]
Courts	2/11 (18.2%)	0/19 (0.0%)	+18.2 [-3.1, 47.7]
Consulted a lawyer	2/11 (18.2%)	1/19 (5.3%)	+12.9 [-10.4, 42.8]
Took no action	7/11 (63.6%)	8/19 (42.1%)	+21.5 [-14.0, 50.0]
Other	0/11 (0.0%)	0/19 (0.0%)	+0.0 [-16.8, 25.9]

Denominators cover only those with a known hukou type who reported at least one rights issue, not all respondents. Multiple responses were allowed; percentages across options may exceed 100%. Negotiating with a boss or HR does not mean a dispute was resolved, and selecting arbitration or court action does not prove that a case was accepted, closed or won.

Records selecting “took no action” alongside a specific action retain a quality flag; the sequence of events is not inferred. Different routes may relate to different incidents or reflect contradictions in answers; this wave cannot reconstruct event-by-event case trajectories.

Because the sample is small, skip logic is incomplete, and actions were not matched individually to events, this section is descriptive. Differences in individual recourse routes are not interpreted as independent causal effects of hukou on rights enforcement. Detailed results retain per-item intervals and raw P values; the narrative avoids selective significance claims.

16 / Barriers Among Those Not Choosing External Procedures

Reason (multiple choice)	Agricultural	Non-agricultural
Fear of losing the job	4/8 (50.0%)	8/16 (50.0%)
Fear of harming future job prospects	3/8 (37.5%)	7/16 (43.8%)
Need the income from this job	4/8 (50.0%)	4/16 (25.0%)
Consider rights enforcement too costly	4/8 (50.0%)	8/16 (50.0%)
Do not know the legal procedures	1/8 (12.5%)	3/16 (18.8%)
Do not believe the issue can be resolved	3/8 (37.5%)	4/16 (25.0%)
Excessive time cost	6/8 (75.0%)	9/16 (56.2%)
Excessive financial cost	3/8 (37.5%)	7/16 (43.8%)
Relationship with the boss is acceptable	1/8 (12.5%)	2/16 (12.5%)
Issue is not serious	0/8 (0.0%)	4/16 (25.0%)
Other	0/8 (0.0%)	0/16 (0.0%)
Did not know rights were violated	0/8 (0.0%)	0/16 (0.0%)

This table includes only respondents who reported a rights issue and did not select labor inspection, labor arbitration or courts. Anyone selecting one of those three routes is excluded. This operational rule is not a complete legal or practical definition of “formal rights enforcement”; legal counsel and union support may also be involved in formal proceedings.

Income dependence, unemployment fears, time costs and financial costs are useful topics for follow-up interviews. Not selecting a reason does not necessarily mean that barrier was absent. The sample is small, reasons may coexist, and the questionnaire did not specify when barriers occurred; population-wide barrier prevalence cannot be inferred.

Future surveys should structure an event module as “problem occurs -> attempt to resolve -> route for each attempt -> resolution -> time and expense -> reason for discontinuing,” with explicit skip logic. Family responsibilities, savings buffers and alternative employment opportunities should be recorded for the same period.

17 / Study Limitations

In this survey, respondents with agricultural hukou reported some rights problems more often. However, the sample is small and participants were not selected randomly. The results are therefore more appropriate for identifying questions for further research than for directly representing all workers.

Source of limitation	Implication for conclusions	Recommended additions
Convenience sampling and self-selection	People more concerned about rights problems may be more inclined to answer; urban group is overconcentrated	Record recruitment channels and coverage; expand rural and migrant-worker samples
Possible workplace/social clustering	Assuming independent responses may underestimate uncertainty	Use anonymized workplace/site IDs; sample and infer by clusters
Recall period vs current status	Current wages/industry may not match circumstances of past events	Standardize past 12 months; record event month and job at the time
Classification error and missingness	Default non-local assignment and vague industry terms may change estimated effects	Separate closed questions; explicit refusal/don't-know flags
Confounding and mediators overlap	Industry and wages may be proxies for causes or pathways of influence	Specify total/direct effect aims before selecting covariates
Small sample and repeated exploration	Wide intervals; individual records may change significance	Expand the sample independently; preregister main outcomes and analysis rules

The absence of clearly detected differences within non-agricultural hukou does not mean age, industry and income play no role. With so few observations, real differences may be missed. Differences in region, occupation and employer conditions may also exist between hukou groups, and this questionnaire cannot eliminate those influences one by one.

Here, “not statistically accurate” means that accurate extrapolation to population prevalence or causal effects cannot be guaranteed; it does not mean that every computation is meaningless. Frequencies, denominators and effect estimates can be independently checked, but this does not remove limitations from sampling representativeness, measurement error or sparse information.

18 / Improving Future Survey Waves

An expanded survey should prioritize greater coverage of people with agricultural hukou and workers explicitly living/working outside their registered locality. Non-agricultural comparison respondents should be recruited in the same regions, industries and types of workplace. The goal is not to discard urban responses until the groups are equally small, but to collect comparison information that is currently missing.

Design element	Recommended implementation
Survey waves	Fix questionnaire versions, recall periods, fieldwork windows and sites; retain recruitment records each wave
Hukou and mobility	Ask separately about hukou type, registered locality, residence, workplace, migration timing and changes in hukou
Employment and treatment	Record industry/job and employer size at time of event, employment arrangement, contract, social-insurance payments, agreed/actual wages and working hours
Rights-related events	For each issue, record when it occurred, frequency, amount involved, injury severity and whether resolved
Potential confounders	Record education, skills, tenure, employer/region, recruitment route and family responsibilities; define adjustment purposes in advance
Longitudinal follow-up	Use optional, random, anonymous follow-up IDs; identify repeat responses and use repeated-measure or cluster methods
Quality assurance	Use logical skip patterns, mutually exclusive “never experienced” checks, missingness reasons, and dual verification of open-ended coding

Sample size: planning illustration only

Suppose a future study used two approximately balanced, independent groups, with assumed prevalence near 50%, two-sided alpha = 0.05 and 80% power, aiming to detect a 15-percentage-point difference. A simplified normal approximation is $n \approx 2 \times 0.5 \times 0.5 \times (1.96 + 0.84)^2 / 0.15^2$, or about 175 participants per group. A formal design should instead use a credible baseline prevalence for its prespecified primary outcome and account for clustering, nonresponse, missingness and stratification needs.

Demonstrating that a difference lies within +/-10 percentage points generally requires more information than detecting a larger difference, and the equivalence bounds must be justified in advance. The illustrative 175 per group cannot simply be reused as the sample size for an equivalence study. If recruitment remains non-probability based, a larger sample reduces some random error but does not automatically remove selection bias.

19 / Conclusions and Data Sources

In this wave, the prevalence of experiencing at least one specific rights issue was 17.2 percentage points higher among agricultural-hukou respondents than among non-agricultural-hukou respondents. Wage arrears and work injury/occupational safety are the differences most worth verifying in further research. Findings depend on a small number of responses, employment status and classification rules, and should be read alongside the full set of sensitivity results.

Point estimates were close for the non-local and local groups under the main rule, but many unclear responses were classified as non-local. Estimates change under the strict rule, so location-related differences cannot be ruled out. Several point estimates within non-agricultural hukou were close, but evidence of equivalence was insufficient, and income groups still showed a substantial sample difference.

Because of sample size, participant selection and resource constraints, these results cannot provide precise statistics for the wider population or determine how much of an employment-treatment gap is caused by hukou. Nevertheless, the questionnaire, database and analytical procedures can continue to be used. With more resources, future waves can expand agricultural-hukou and non-local worker samples, improve questions and test whether the differences persist.

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